Microfluidic Transport in Porous Media

Department of Mechanical Engineering • Fluid Dynamics Review

1. Darcy Flow Regime

Flow through a packed porous bed satisfies Darcy's law when the Reynolds number based on mean grain diameter remains well below unity (Re < 1). The macroscopic superficial velocity vector q depends on hydraulic permeability kappa and dynamic fluid viscosity mu:

$$ \vec{q} = -\frac{\kappa}{\mu} \nabla p $$

In the presence of gravitational acceleration g along the vertical axis, the pressure gradient is augmented by hydrostatic head:

$$ \vec{q} = -\frac{\kappa}{\mu} (\nabla p - \rho \vec{g}) $$

2. Permeability Characterization

The Kozeny-Carman empirical equation estimates permeability from bed porosity phi and specific surface area per unit volume S_v:

$$ \kappa = \frac{\phi^3}{c_0 S_v^2 (1 - \phi)^2} $

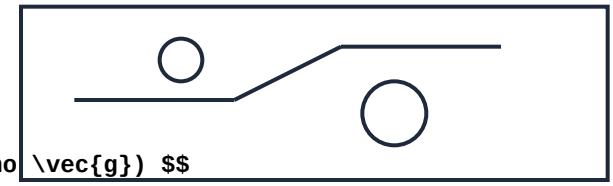

Figure 1: Tortuous pore network model.